

UNIT: 5 Control systems

#. OPEN AND CLOSED LOOP SYSTEM:

1. Open loop system:

→ A system in which output is dependent on the input but controlling action or input is totally independent of the output of the system, is called an open loop system.

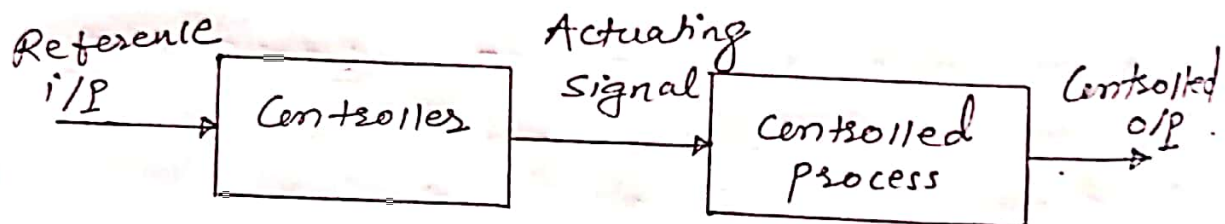


fig: open loop system.

→ There are 2-elements of open loop system:

- (a) Controller
- (b) Controlled process.

(a) Controller:

- An i/p signal or command is applied to the controller.

(b) Controlled process:

- An i/p signal is applied to the Controller, whose o/p acts as the actuating signal then controls the controlled process so that the controlled output is obtained.

- The Controller can be an amplifier, mechanical linkage, filter or other control element, depending on the nature of the system.

Example: Light switch, Volume of stereo system, Automatic washing machine, Automatic Tea/coffee maker.

Advantages:

- i> Simple in construction.
- ii> Easy to maintain.
- iii> Simple to design & economical.
- iv> Convenient when o/p is difficult to measure.

Dis-Advantages:

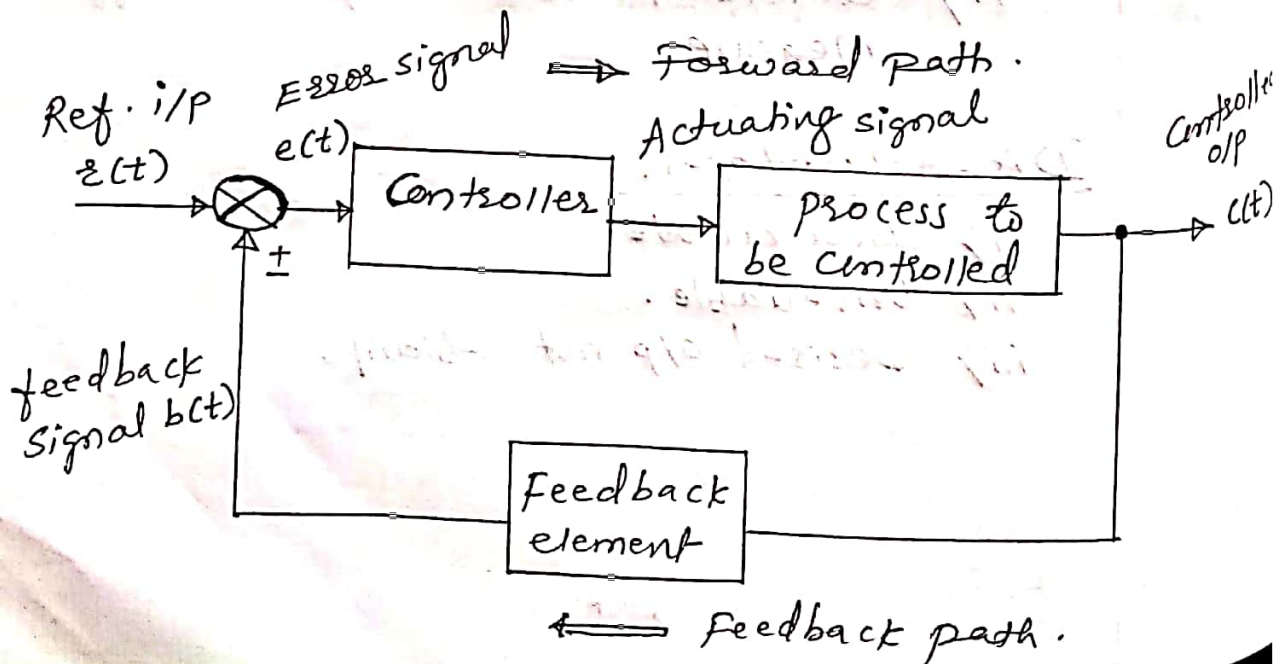
- i> Inaccurate.
- ii> Unreliable.
- iii> Desired o/p not obtain.

2. Closed loop system :

→ A system in which the Controlling action or input is somehow dependent on the output or changes in output, is called closed loop system.

- In this case the Controlled output is feedback through a feedback element and compared with the reference input. Thus the actuating signal is the difference of desired output and reference input.

FEED BACK : Feedback is a property of the system by which it permits the output to be compared with the reference input so that appropriate controlling action can be decided.



- Initially Reference i/p $[r(t)]$ is applied.
- The part of o/p, which is to be decided is fed back by feedback element to the reference i/p.
- The signal which is o/p of feedback element is called 'feedback signal' $b(t)$
- It is then compared with the reference i/p giving error signal $e(t) = r(t) \pm b(t)$.
- This error signal is then modified by Controller and decides the proportional actuating signal for the process to be controlled.
- This modification is such that error will approach to zero. This signal then actuates the actual system and produces an output.
- As output is controlled one, hence called Controlled output $c(t)$.

Advantages :

- i) High Accuracy
- ii) Large range of Bandwidth. (Fast response)
- iii) Reduced effect of nonlinearities and distortions.
- iv) Can stabilize an open-loop unstable plant.

Dis-advantages :

- i) more Complex.
- ii) Risk of instability.

#. Modeling of Mechanical Systems using Spring, Dashpot and Mass equivalence:

- Based on Type of motion mechanical systems are classified into two types:

- i) Translational Mechanical systems.
- ii) Rotational Mechanical systems.

✓ (i) Translational mechanical systems:

- (a) Translational mechanical systems move along a straight line.
- (b) These systems mainly consist of three basic elements. Those are Mass, Spring and dashpot or damper.
- (c) If a force is applied to a translational mechanical system, then it is opposed by opposing forces due to mass, elasticity and friction of the system.

④ since the applied force and the opposing forces are in opposite directions, the algebraic sum of the forces acting on the system is zero.

#. Basic elements of T.M.S :

1. Springs:

— spring is an element, which stores potential energy. If a force is applied on spring 'k', then it is opposed by an opposing force due to elasticity of spring. This opposing force is proportional to the displacement of the spring 'x'.



$$F_k \propto x$$

$$F_k = kx$$

$$\therefore F = f_k = kx$$

$$\therefore \boxed{F = kx} \quad - \quad (1)$$

Where , F = Applied force
 k = Constant f
 x = displacement.

F_k = opposing force by elasticity

- Eqn (1) indicates that as per Newton's third law, the force F is equal in size and in the opposite direction to the force exerted by the stretched spring.

*. Energy stored when there is extension x ,

$$E = \frac{1}{2} kx^2 = \frac{1}{2} \frac{F^2}{k} \quad (\because F = kx)$$

2. Dashpots:

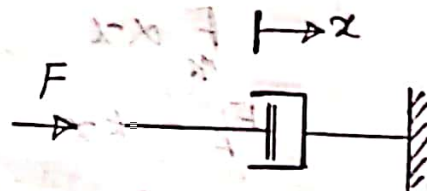
- A dashpot is a mechanical device (damper) which resists motion via viscous friction.

- The resulting force is proportional to velocity but acts in opposite direction. It is used in doors & shock absorbers.

$$F_b \propto v$$

$$\therefore F_b = cv$$

$$F_b = c \cdot \frac{dx}{dt}$$



Dashpot.

$$\therefore \boxed{F = F_b = c \cdot \frac{dx}{dt}} \quad \text{---} \quad (2)$$

Where: F = Applied force

F_b = opposing force due to friction.

C = Constant

v = velocity of the piston.

- In a dashpot no energy is stored. It does not return to its original position when there is no force input. The dashpot dissipates energy rather than storing it.

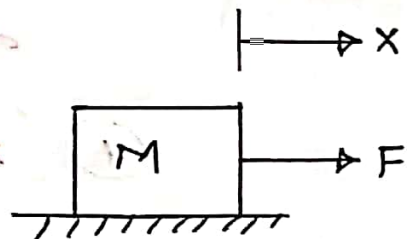
*. power dissipated, $P = Cv^2$.

3. Masses:

- The masses represent the inertia or resistance to acceleration. If a force F is applied on a body having mass M , then it is opposed by an opposing force due to mass. This opposing force is proportional to the acceleration of the body. Assume elasticity & friction are negligible. As per Newton's Law: (IInd)

$$F_m \propto a$$

$$\therefore F_m = Ma = M \cdot \frac{d^2x}{dt^2}$$



$$\therefore \boxed{F = F_m = M \cdot \frac{d^2x}{dt^2}} \quad - (3)$$

Where; x = displacement

M = mass

F_m = opposing force by mass.

F = applied force.

*. "Energy (kinetic energy) stored" in the mass when it is moving with velocity v , & released when it stops moving,

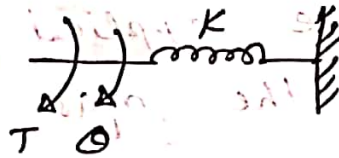
$$E = \frac{1}{2}mv^2$$

vii) Rotational Mechanical Systems:

- Basic elements of Rotational Mech. Systems are:

- ① Torsional spring,
- ② Rotary damper,
- and ③ Moment of inertia.

① Torsional spring:



- In a torsional spring the angle rotated (θ) is proportional to the torque (T),

$$\therefore T_k \propto \theta$$

$$\therefore T_k = k\theta$$

$$T = T_k = k\theta.$$

$$\boxed{T = k\theta} \quad \text{--- (1)}$$

where;

T = Applied Torque.

T_k = Resistive Torque.

θ = angular displacement.

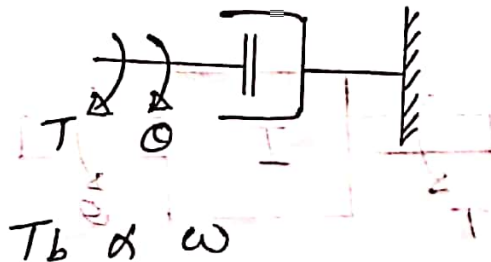
k = Stiffness of spring.

* "Energy stored" by the torsional spring when twisted through an angle θ ,

$$E = \frac{1}{2} k \theta^2 = \frac{1}{2} \cdot \frac{T^2}{k} \quad (\because T = k\theta)$$

(2) Rotary dampers:

- In Rotary dampers, a disc is rotated in a fluid and the resistive torque (T_b) is proportional to the angular velocity (ω).



$$\therefore T_b = C \omega$$

$$T_b = C \cdot \frac{d\theta}{dt}$$

$$\therefore T = T_b = C \cdot \frac{d\theta}{dt}$$

$$\therefore \boxed{T = C \cdot \frac{d\theta}{dt}} \quad - (2)$$

Where;

T = Applied torque.

T_b = Resistive torque

C = Constant.

θ = Angular displacement.

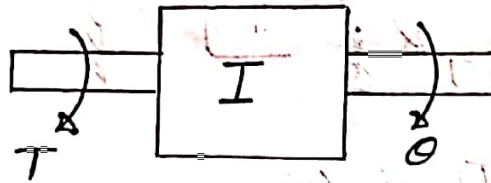
ω = Angular velocity.

* "Power dissipated" by the rotary damper when rotating with an angular velocity ω ,

$$P = C\omega^2$$

③. Moment of inertia (I): -

- Greater the M.I. the greater the torque (T) needed to produce an angular accⁿ α .



$$T_I \propto \alpha$$

$$\therefore T_I = I \cdot \alpha$$

$$T_I = I \cdot \frac{d^2\Theta}{dt^2}$$

$$\therefore T = T_I = I \cdot \frac{d^2\Theta}{dt^2}$$

$$\therefore \boxed{T = I \cdot \frac{d^2\Theta}{dt^2}} \quad - \quad \text{③}$$

Where;

I = moment of inertia.

T = Applied torque

T_I = Resistive torque

α = angular acceleration.

Θ = angular displacement

*. "Energy stored" by the mass rotating with
an angular velocity, ω
$$= \frac{1}{2} I \omega^2$$